AI audio software hands-on session

Making Speech Clearer

Streaming Neural Speech Enhancement in Practice (FastEnhancer)

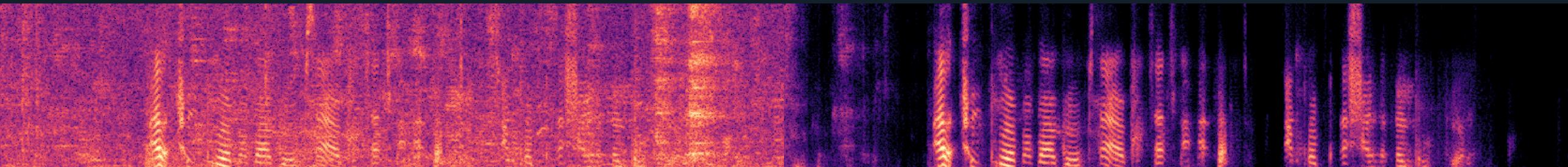
Hsin-Te Hwang 黃信德　Former Postdoctoral Researcher, IIS, Academia Sinica
Assoc. Prof. Chen 陳雅淇　Chung Yuan Christian University

Tue 20 Oct 2026, 12:10–13:00　Lecture Hall 112, Teaching Building, CYCU

github.com/aask1357/fastenhancer

Scan for all course materials
Web • Colab • Slides

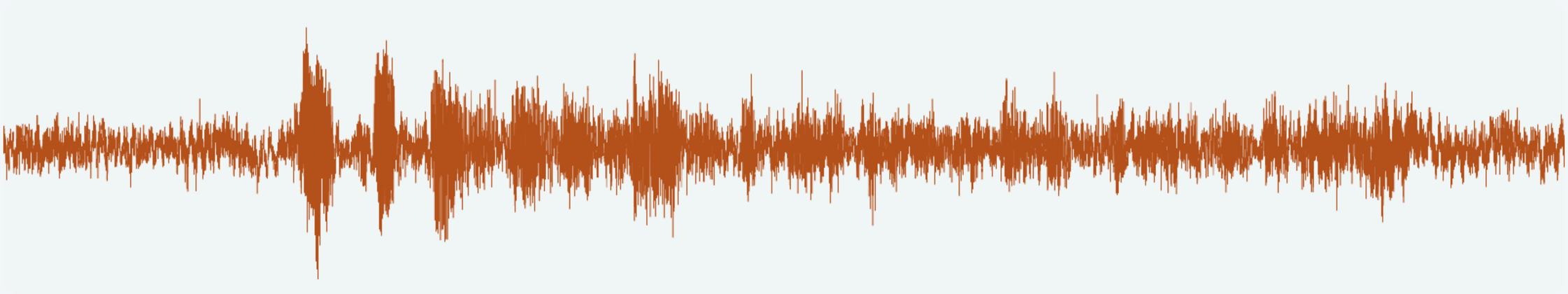
Noisy (café)　　Denoised



Today's 50 minutes

Time	Part	What you will do
0–4 min	Opening demo	Hear the difference denoising makes
4–12 min	Concepts	Spectrograms, masks, streaming, speed and latency
12–30 min	Colab	Run it cell by cell and see how the model works
30–38 min	Live denoising	The model running live on your own device
38–45 min	Challenge	Find the noises it cannot remove
45–50 min	Wrap-up	Applications and what to take home

Sound is a long list of numbers

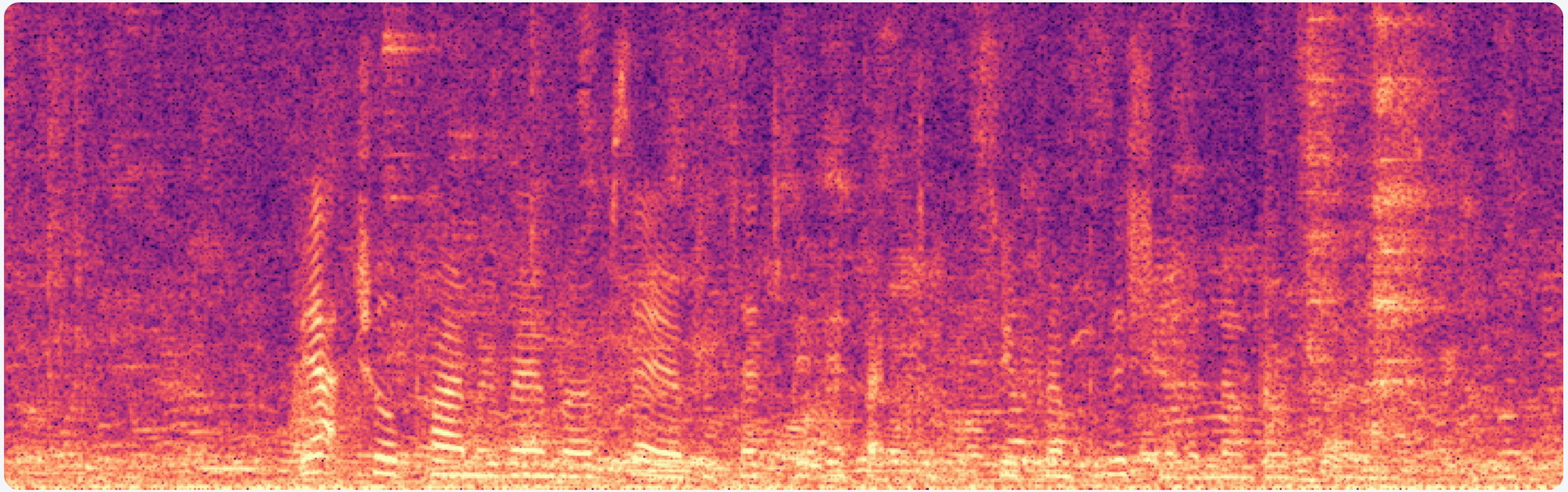


16,000

numbers every second

The microphone measures the air vibration 16,000 times a second: a **16 kHz sampling rate**.

Another view: the spectrogram



Time across, frequency up
Low pitch at the bottom, high at the top; brighter is louder

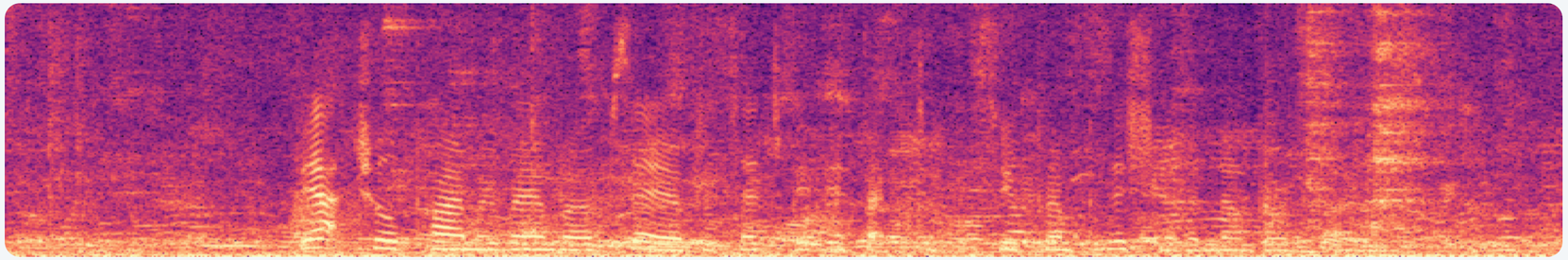
Noise: a fog
Café chatter, cups and plates

Speech: stripes
Harmonics from the vibrating vocal folds

Denoising = laying a mask on top

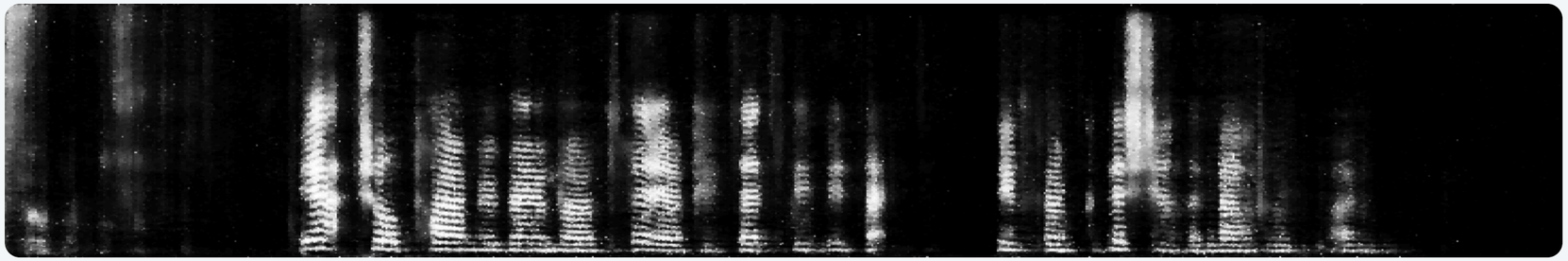
Noisy

What the microphone hears



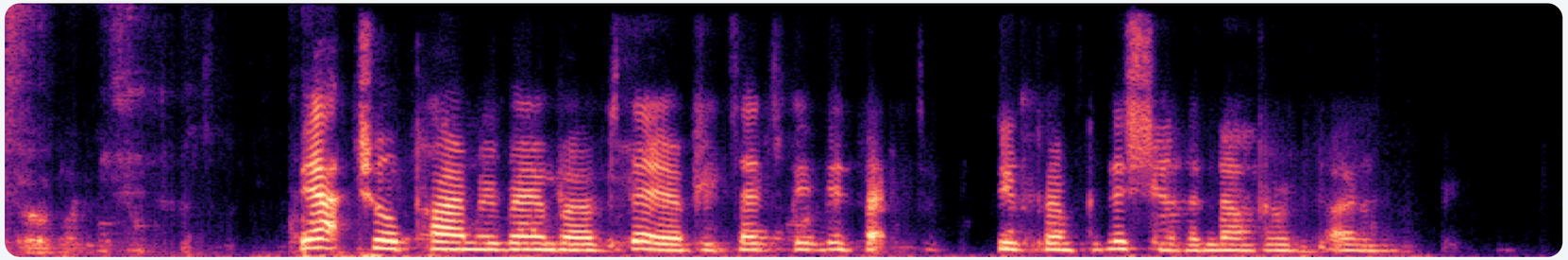
× Mask

White keeps, black removes

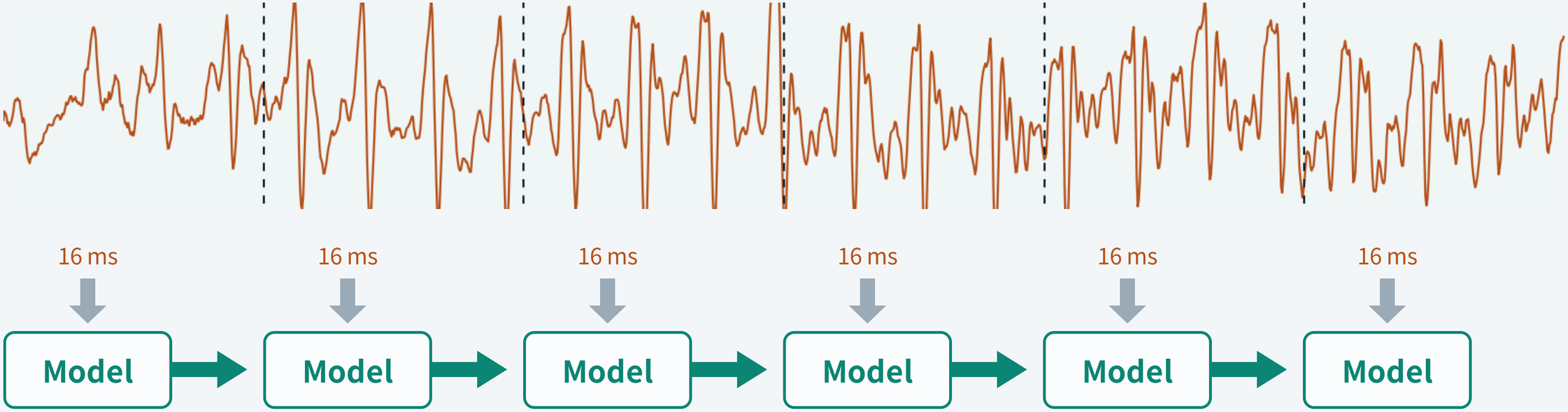


= Denoised

The fog clears, the stripes stay



Streaming: just 16 ms at a time



A real-time system never gets the whole sentence; it must process while it listens. Every 256 numbers (16 ms) go into the model; **the green arrows are the "memory"**, passing on what it has heard.

Fast enough? Faster than the sound itself

16 ms → **~1 ms**

of audio

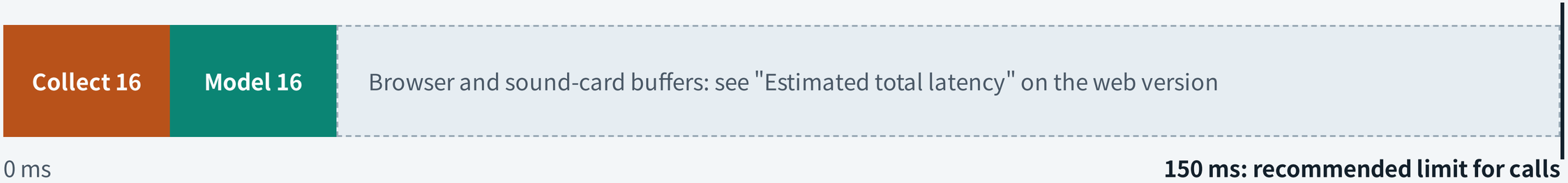
to compute (Base model, one CPU core)

RTF = compute time ÷ audio length Below 1 means real time

Model	File size	Server RTF	Phone RTF
Tiny	122 KB	0.038	0.113
Base	404 KB	0.059	0.228
Small	813 KB	0.083	0.293

Server: Linux, one CPU core, Python on audio files. Phone: Galaxy A55, live streaming in the web version's "Live microphone" mode.

Fast, but still a little late



32 ms

The model's own latency: collect 16 ms, then the output is another 16 ms late

< 150 ms

ITU-T G.114 recommends at most 150 ms one-way delay for calls

ON YOUR LAPTOP

Open Colab, press ▶ cell by cell

Scan the QR code, or press Open in Colab in the course repo:
github.com/tehan0530/CYCU-fastenhancer-course

1. If you see "This notebook was not authored by Google", choose "Run anyway"
2. Start at section 0 and press ▶ on the left of each cell
3. No need to change the code; follow along and watch the results

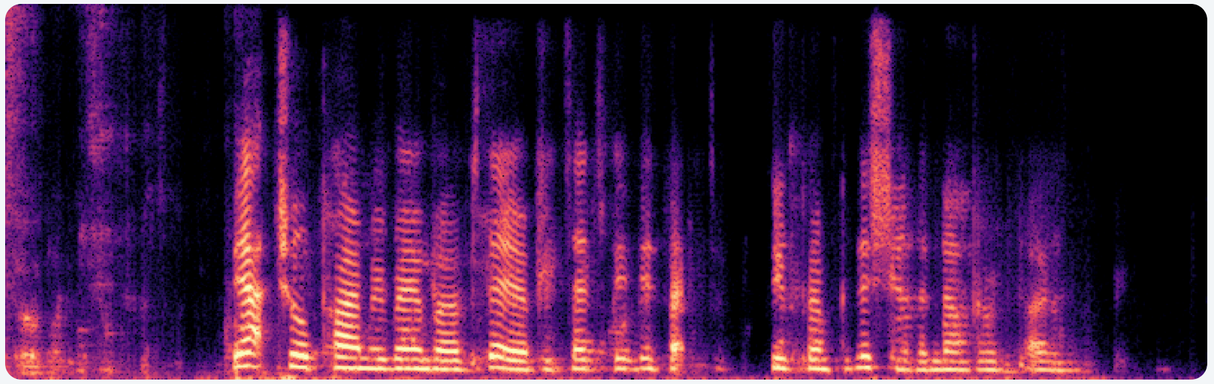
Needs a Google account. No account, or cannot sign in? Open the "pre-run copy" linked in the repo README; no sign-in needed.

Guess: what if we remove the memory?

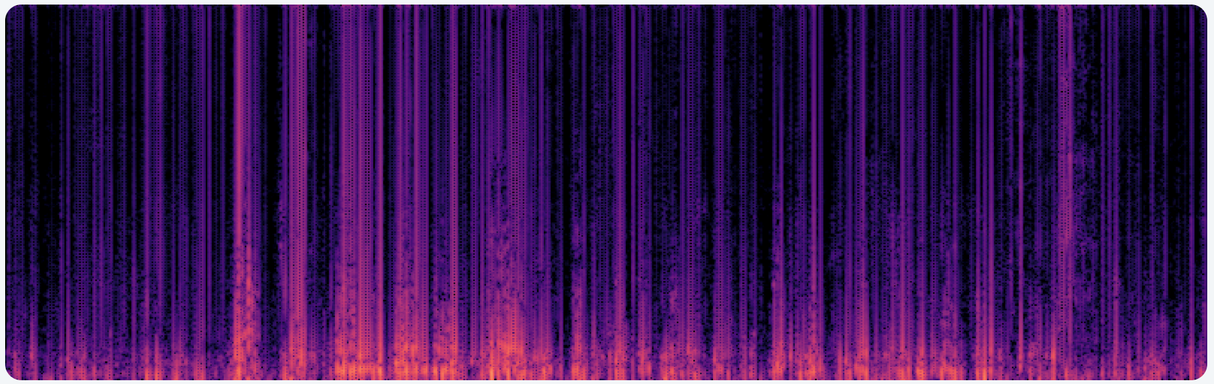
The memory is wiped every 16 ms, so the model sees only the frame in front of it.

From 16 ms alone it cannot tell speech from noise, nor rebuild the sound. Memory lets it reason: "this hum has been here all along, so it is noise."

With memory (normal)



No memory



PHONE, TABLET OR LAPTOP

The same model, live on your device

benevolent-pithivier-3ffb14.netlify.app

The streaming loop you just ran cell by cell in Colab is exactly what the web page does.

Nothing to install; the model runs on your device and your audio is never uploaded. Switch to EN at the top right.



CHALLENGE • 7 MIN

Find the noises it cannot remove

Noise	Removed?	Why do you think so?
Fan, air conditioner		
Clapping, typing		
Crinkling a plastic bag		
Background music with singing		
Two people talking at once		

Watch the spectrograms in "Live microphone", or use "Record 5 s" and compare

Keep playing at home

Web version

benevolent-pithivier-3ffb14.netlify.app

Live denoising on your phone

Colab notebook

github.com/tehan0530/CYCU-fastenhancer-course

Press Open in Colab in the README; every cell from today is there

Code and paper

github.com/aask1357/fastenhancer

Start here to train your own model

Denoising is all around you

Video calls • phone calls • hearing aids and cochlear implants • electrolarynx speech for people who have lost their voice box

Model: FastEnhancer (Ahn et al., 2025, arXiv:2509.21867, MIT License) Example clips: VCTK (CC BY 4.0) + DEMAND (CC BY-SA 3.0)

APPENDIX

Host your own web version

The web version is just a bundle of files: HTML, the model and the runtime. The model runs on the visitor's device, so no server is needed; any place that gives you an https address works.

Option	Effort	Works on phones?	Address
Netlify Drop	One drag	Yes	xxx.netlify.app
Your own computer	One command	No, this computer only	localhost
GitHub Pages	A few settings	Yes	account.github.io

All files: github.com/tehan0530/CYCU-fastenhancer-course

Three options, step by step

Netlify Drop

1. On the repo page press Code → Download ZIP, then unzip
2. Sign up and log in to Netlify (free)
3. Drag the web folder onto app.netlify.com/drop

Without logging in, the site is deleted after about an hour

Your own computer

1. Install Python 3 and download the repo
2. In the repo folder run:
   python tools/serve.py
3. Open localhost:8765

Don't use python -m http.server; it may fail to load on Windows

GitHub Pages

1. Fork this repo
2. Settings → Pages → Source: GitHub Actions
3. Enable workflows on the Actions tab and run Deploy web to Pages

Address: account.github.io/repo-name